

TANVIR HOSSAIN

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SUMMARY

Security and privacy researcher (Ph.D. expected 2026) working across the full hardware/software stack, with depth in hardware and computer-architecture security and in applied-cryptographic side-channel analysis. Author of 17 peer-reviewed publications — including IACR TCHES, ACM TODAES, IEEE/ACM ICCAD, and the Journal of Hardware and Systems Security — spanning hardware Trojan attacks and defenses, side-channel and fault analysis of cryptographic primitives, ML-assisted security, and compiler-based (LLVM) software protections. Champion of the IEEE HOST 2023 Microelectronics Security Challenge. Experienced experimental researcher, collaborator across institutions, and presenter; proficient in C/C++, Python, and Verilog. Motivated to co-design trustworthy AI systems with strong, verifiable security and privacy guarantees.

EDUCATION

Ph.D. in Electrical Engineering (Hardware Security), The University of Kansas Expected 2026

Advisor: Dr. Tamzidul Hoque. Dissertation: security across the hardware/software stack — Trojan-assisted attacks on design-for-security primitives, side-channel and fault analysis, and ML-assisted vulnerability detection.

M.S. in Electrical Engineering, The University of Kansas May 2025

B.Sc. in Electrical and Electronic Engineering, Ahsanullah University of Science and Technology 2016 – 2021

RESEARCH AREAS & TECHNICAL SKILLS

Research Areas:	Hardware and computer-architecture security; trustworthy AI / ML-system security; side-channel and fault analysis and countermeasures; applied cryptography; machine learning for security.
Comp. Architecture & Systems:	RTL/ASIC design and verification; SoC and FPGA platforms; microarchitectural side-channel analysis; gem5, McPAT, Intel VTune; embedded/COTS processor security.
Applied Crypto & Side-Channel:	Power/EM side-channel analysis of cryptographic primitives (e.g. ASCON), fault injection, and countermeasure evaluation; residue-number-coding data protection.
AI/ML for Security:	ML-assisted side-channel analysis; golden-chip-free hardware Trojan detection and localization; model-guided fault-injection optimization.
Languages & Compilers:	C, C++, Python, Verilog, VHDL, Assembly (MIPS), MATLAB, Tcl, Bash; LLVM compiler-pass development.
EDA & Lab Tooling:	Cadence Genus, Encounter, Virtuoso; Xilinx Vivado, ISE; Intel Quartus; ModelSim; HSPICE, PSpice; ChipWhisperer Husky; Keysight/Riscure Inspector, Spyder, Pinata; EM probes; oscilloscopes.

RESEARCH EXPERIENCE

Keysight Technologies May 2026 – Aug 2026
Research Intern — Machine Learning Security (Applied ML & Fault Injection) Santa Rosa, CA

- Researching model-guided optimization and adaptive-learning methods that automatically tune fault-injection parameters to improve vulnerability-detection coverage over baseline sweep-based flows.
- Designing experiments and telemetry instrumentation, scalable hyperparameter search, and reliability metrics; delivering a prototype demonstration on production device-security test tooling.

The University of Kansas Jan 2022 – Present
Graduate Research Assistant, Hardware Security (Advisor: Dr. Tamzidul Hoque) Lawrence, KS

- **Hardware/architecture security:** designed Trojan-assisted meta-attacks that systematically locate and neutralize design-for-security primitives, demonstrating authentication bypass of ring-oscillator PUFs and unlocking of dynamically obfuscated scan chains (ACM TODAES 2026).
- **Applied cryptography:** led a security analysis of lightweight homomorphic obfuscation schemes that protect software secrets against hardware Trojans, characterizing their guarantees and limits (IACR TCHES 2026).
- **Software/PL-layer protection:** built HOACS, a compiler-based (LLVM) and C-library framework that keeps secrets encoded during execution on untrusted COTS processors via residue number coding — protecting cryptographic and arithmetic workloads without hardware changes.
- **ML for security:** developed golden-chip-free, ML-assisted power side-channel methods for detecting and localizing hardware Trojans on FPGAs (Journal of Hardware and Systems Security 2026; IEEE NAECON 2023).

- **Tooling & experimental R&D:** built SPHINX, an automated framework that extracts security primitives from gate-level netlists; proposed a gray-box, pre-deployment evaluation methodology for opaque COTS hardware using power/EM side channels; demonstrated power side-channel and fault attacks on the ASCON standard and evaluated countermeasures.
- **Collaboration & communication:** co-authored with teams at KU, the University of Florida, and Wright State University; mentored four NSF REU undergraduates across two institutions; co-taught a hands-on hardware course and delivered side-channel tutorials and research talks.

PUBLICATIONS

17 peer-reviewed publications (7 journal articles, 10 conference papers). [Google Scholar](#).

Journal Articles

1. **Tanvir Hossain**, Matthew Showers, Mahmudul Hasan, and Tamzidul Hoque, “On the Security of Lightweight Homomorphic Obfuscation for Protecting Against Hardware Trojans,” *IACR Transactions on Cryptographic Hardware and Embedded Systems (TCHES)*, vol. 2026, no. 2, pp. 736–763, 2026. DOI: [10.46586/tches.v2026.i2.736-763](#)
2. **Tanvir Hossain**, S. M. Mojahidul Ahsan, Prabuddha Chakraborty, Swarup Bhunia, and Tamzidul Hoque, “Attacking the Protections of Hardware using Trojan-Assisted Meta Attacks,” *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 2026. DOI: [10.1145/3804452](#)
3. **Tanvir Hossain**, Ashutosh Ghimire (joint first author), Fathi Amsaad, Tamzidul Hoque, and Ahmed Sherif, “AI-Assisted Side-Channel Analysis for Golden-Chip Free Detection and Localization of Hardware Trojans,” *Journal of Hardware and Systems Security*, 2026. DOI: [10.1007/s41635-026-00182-4](#)
4. Sriram Prabhakara Rao, Shree Ram Abayankar Balaji, Farishta Rahman, Tasneem Suha, Tanzim Mahfuz, **Tanvir Hossain**, Mahmudul Hasan, Md Sakib Hasan, Tamzidul Hoque, Prabuddha Chakraborty, and Prakash Ranganathan, “Secure and Trustworthy Microelectronics: Vulnerabilities, Solutions, and Trends,” *Journal of Hardware and Systems Security*, vol. 9, pp. 163–189, 2025. DOI: [10.1007/s41635-025-00165-x](#)
5. Md Mashfiq Rizvee, Fairuz Shadmani Shishir, **Tanvir Hossain**, Tamzidul Hoque, Domenic Forte, and Sumaiya Shomaji, “A Persistent Hierarchical Bloom Filter-based Framework for Scalable Authentication and Tracking of ICs,” *ACM Journal on Emerging Technologies in Computing Systems*, vol. 21, no. 4, pp. 1–22, 2025. DOI: [10.1145/3748650](#)
6. **Tanvir Hossain** and Md Istiaque Rahaman, “Band gap tuning and variable quantum barrier of armchair graphene nanoribbons by inducing antidote topologies,” *Journal of Computational Electronics*, vol. 23, no. 1, pp. 32–41, 2024. DOI: [10.1007/s10825-023-02108-7](#)
7. **Tanvir Hossain**, Md Istiaque Rahaman, and Mahbub Alam, “Antidote-induced armchair graphene nanoribbon based resonant tunneling diodes,” *Semiconductor Science and Technology*, vol. 36, no. 7, art. 075023, 2021. DOI: [10.1088/1361-6641/ac01fc](#)

Conference Proceedings

1. **Tanvir Hossain**, S. M. Mojahidul Ahsan, and Tamzidul Hoque, “SPHINX: A Framework for Security Primitive Hardware Identification and Extraction,” *Great Lakes Symposium on VLSI (GLSVLSI)*, 2026. DOI: [10.1145/3787109.3815307](#)
2. S. M. Mojahidul Ahsan, Muhammad Sakib Shahriar, Mrittika Chowdhury, **Tanvir Hossain**, Md Sakib Hasan, and Tamzidul Hoque, “Accurate, Yet Scalable: A SPICE-based Design and Optimization Framework for eNVM-based Analog In-memory Computing,” *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, pp. 1–9, 2024. DOI: [10.1145/3676536.3676827](#)
3. Ashutosh Ghimire, Fathi Amsaad, **Tanvir Hossain**, Tamzidul Hoque, and Ahmed Sherif, “FPGA Hardware Trojan Detection: Golden-Free Machine Learning Approach,” *IEEE National Aerospace and Electronics Conference (NAECON)*, pp. 181–186, 2023. DOI: [10.1109/NAECON58068.2023.10365812](#)
4. **Tanvir Hossain**, S. M. Mojahidul Ahsan, and Tamzidul Hoque, “Potential and Pitfalls of Multi-valued Logic Circuits for Hardware Security,” *IEEE Dallas Circuits and Systems Conference (DCAS)*, pp. 1–6, 2023. DOI: [10.1109/DCAS57389.2023.10130261](#)
5. S. M. Mojahidul Ahsan, **Tanvir Hossain**, Md Sakib Hasan, and Tamzidul Hoque, “Resistive RAM-based PUF: Challenges and Opportunities,” *IEEE Dallas Circuits and Systems Conference (DCAS)*, pp. 1–6, 2023. DOI: [10.1109/DCAS57389.2023.10130179](#)
6. Fairuz Shadmani Shishir, Md Mashfiq Rizvee, **Tanvir Hossain**, Tamzidul Hoque, Domenic Forte, and Sumaiya Shomaji, “A Persistent Hierarchical Bloom Filter-based Framework for Authentication and Tracking of ICs,” *3rd IEEE International Workshop on Silicon Lifecycle Management*, 2024. DOI: [10.48550/arXiv.2408.16950](#)
7. **Tanvir Hossain**, Md Shyeem Rahman, Md Mushfiqur Rahman, and Ahmed Rabbi Arini Dibbo, “Performance Analysis of Graphene Nanoribbon Field Effect Transistor (GNRFET) based 6T and 7T SRAMs,” *23rd International Conference on*

Computer and Information Technology (ICCIT), pp. 1–5, 2020. DOI: [10.1109/ICCIT51783.2020.9392713](https://doi.org/10.1109/ICCIT51783.2020.9392713) (**Best Paper Award**)

8. Andrea Ramirez-Salgado, Swarup Bhunia, Pavlo Antonenko, et al. (incl. **T. Hossain**), “NSF IUSE: Empowering Future Engineers — An Inclusive Curriculum for AIoT and Intelligent Embedded Systems,” *ASEE Annual Conference & Exposition*, 2025. DOI: [10.18260/1-2--55641](https://doi.org/10.18260/1-2--55641)
9. Andrea Ramirez-Salgado, **Tanvir Hossain**, Swarup Bhunia, and Pavlo Antonenko, “Supporting Hardware Engineering Career Choice in First-Year Engineering Students,” *ASEE Annual Conference & Exposition*, 2024. DOI: [10.18260/1-2--46979](https://doi.org/10.18260/1-2--46979)
10. Andrea Ramirez-Salgado, **Tanvir Hossain**, Tamzidul Hoque, Swarup Bhunia, et al., “Engaging Students in Exploring Computer Hardware Fundamentals Using FPGA Board Games,” *ASEE Annual Conference & Exposition*, 2023. DOI: [10.18260/1-2--42715](https://doi.org/10.18260/1-2--42715)

SELECTED TALKS, MENTORING & PROFESSIONAL SERVICE

Invited Talks & Tutorials	<i>Leakage Unplugged: A Hands-On Tutorial on Power Side-Channel Attacks</i> — I2S Student Research Symposium, 2025. <i>Power Side-Channel and Fault Attacks on ASCON and their Countermeasures</i> — Information Advantage Scholars Visit, I2S, University of Kansas, 2024.
Mentoring	Mentored four undergraduate researchers in the NSF REU program (University of Kansas / Wright State University).
Journal Reviewer	IEEE Embedded Systems Letters; Journal of Hardware and Systems Security; SN Computer Science; Physica Scripta; Journal of Physics: Condensed Matter.
Conference / TPC	Technical Program Committee, IEEE Computer Society Annual Symposium on VLSI (ISVLSI 2026); Reviewer, ASEE Annual Conference (2024, 2026).

SELECTED HONORS & AWARDS

MAY 2023	Champion, Microelectronics Security Challenge (IP Security) — IEEE International Symposium on Hardware Oriented Security and Trust (HOST 2023).
2022, '23, '26	NSF Travel Grants — IEEE HOST Symposium (three times).
JANUARY 2024	First Place, Poster Presentation — I2S Student Research Symposium, University of Kansas.
APRIL 2023	David D. and Mildred H. Robb Award — EECS Department, University of Kansas.
APRIL 2023	People’s Choice & Third Place, Poster — KU Graduate Engineering Association Research Showcase.
MARCH 2021	Best Paper Award — 23rd International Conference on Computer and Information Technology (ICCIT 2020).